# Freight Rate Prediction Assessment

## Validation approach and model performance

Model: LightGBM absolute-error regression in a scikit-learn Pipeline. The pipeline imputes missing numeric values, one-hot encodes pickup, delivery, and equipment with unknown-category handling, and engineers calendar and coordinate-delta features. The load identifier is excluded.

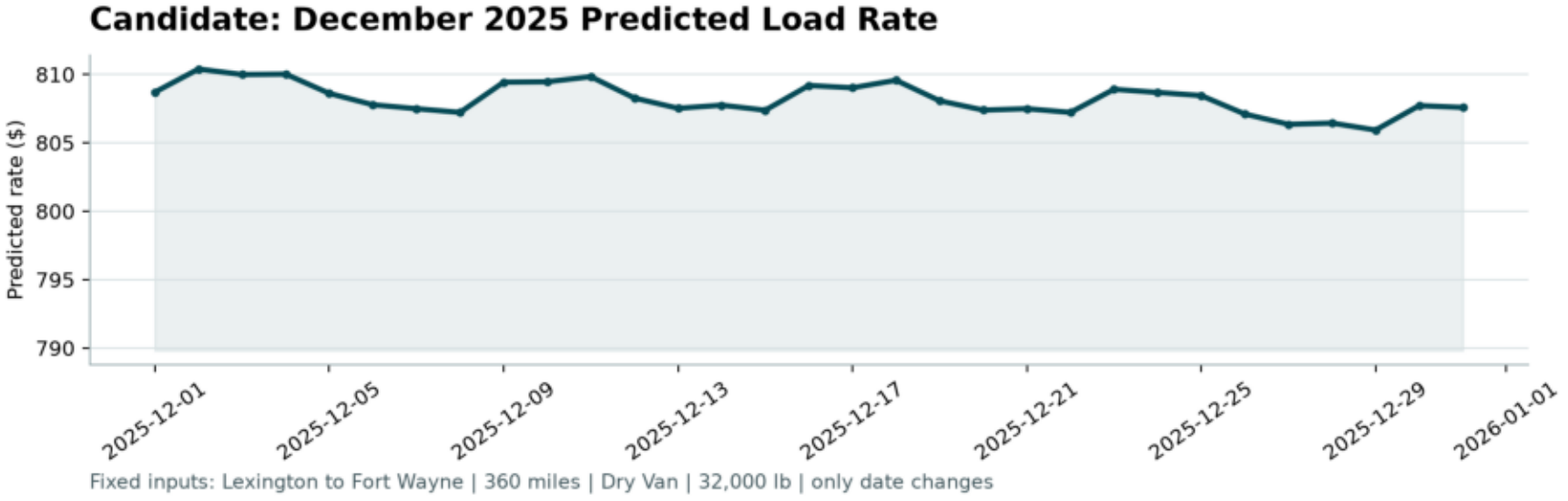
Split: chronological holdout. Train rows: 38,477 (through 2025-08-31); holdout rows: 9,523 (2025-09-01 to 2025-10-31). This holds out the last two complete months to approximate future-month prediction and prevent random temporal leakage.

| Metric | Result |
|---|---|
| Holdout MAE | $117.88 |
| Holdout RMSE | $636.64 |
| Holdout $R^2$ | 0.8260 |
| Holdout MAPE | 5.09% |
| Mileage-only baseline MAE | $256.95 |
| MAE improvement vs baseline | 54.1% |

Data checks: 300 training weights and 374 market-index values are missing and imputed within the training pipeline. 292 weights are negative; they are retained as observed inputs rather than silently altered.

Final outputs: 12,000 validation predictions and 31 December chart predictions. For December, market signals and coordinates are absent from the supplied inputs; the same pipeline handles them using training-derived median imputation.

Validation metrics are from the held-out labeled training data; no final validation labels were used.


Candidate: December 2025 Predicted Load Rate
Predicted rate ($)
810
805
800
795
790
2025-12-01
2025-12-05
2025-12-09
2025-12-13
2025-12-17
2025-12-21
2025-12-25
2025-12-29
2026-01-01
Fixed inputs: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb | only date changes